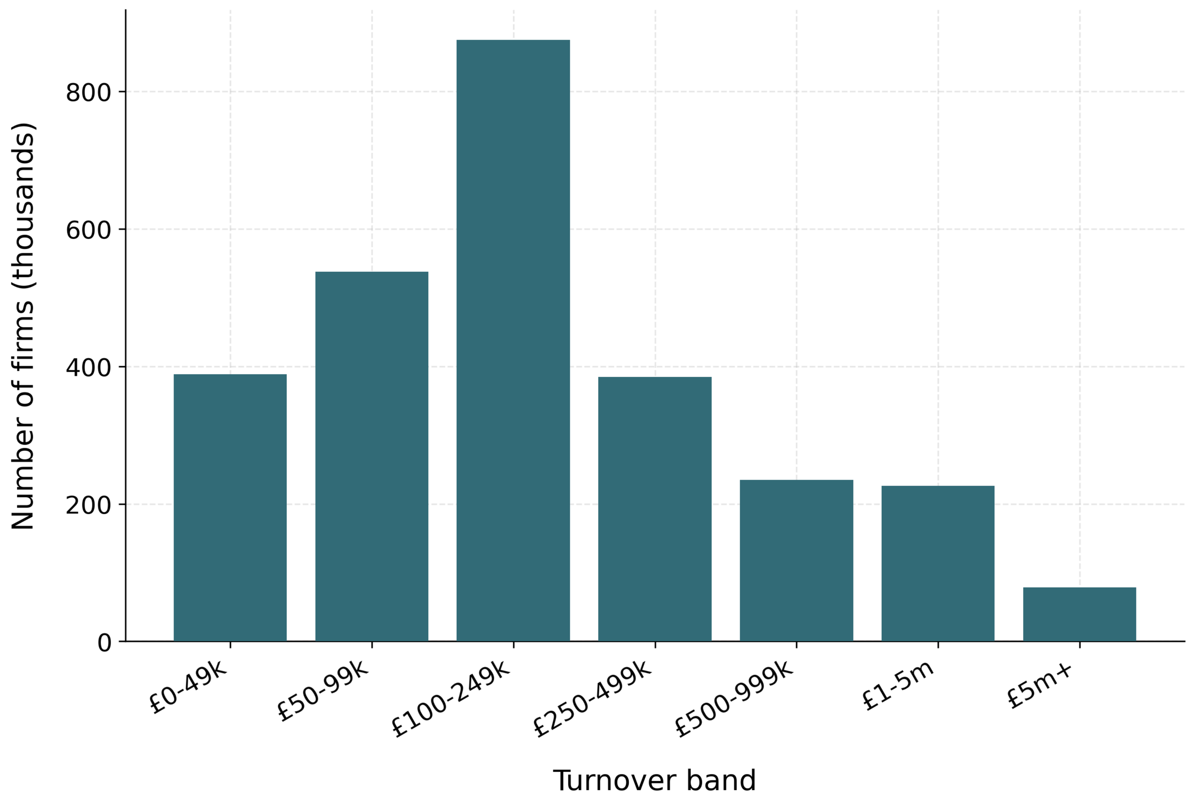
A Firm-Level Microsimulation for VAT Policy Analysis

Vahid Ahmadi

PolicyEngine | Open, reproducible firm-level VAT microsimulation | `vahid@policyengine.org`

The problem: VAT as a tax *notch*

The UK VAT registration threshold is a notch, not a kink. Once taxable turnover crosses T^*, 20% VAT falls due on the *whole* turnover, not just the increment — a discrete liability jump of $\tau T^* = \pounds17{,}000$ at $T^* = \pounds85{,}000$. Threshold £85,000 (2023–24 data year), raised to £90,000 in April 2024 — among the highest in the OECD — and it sits in a *dense* part of the firm-size distribution.



Three reform families, one basis

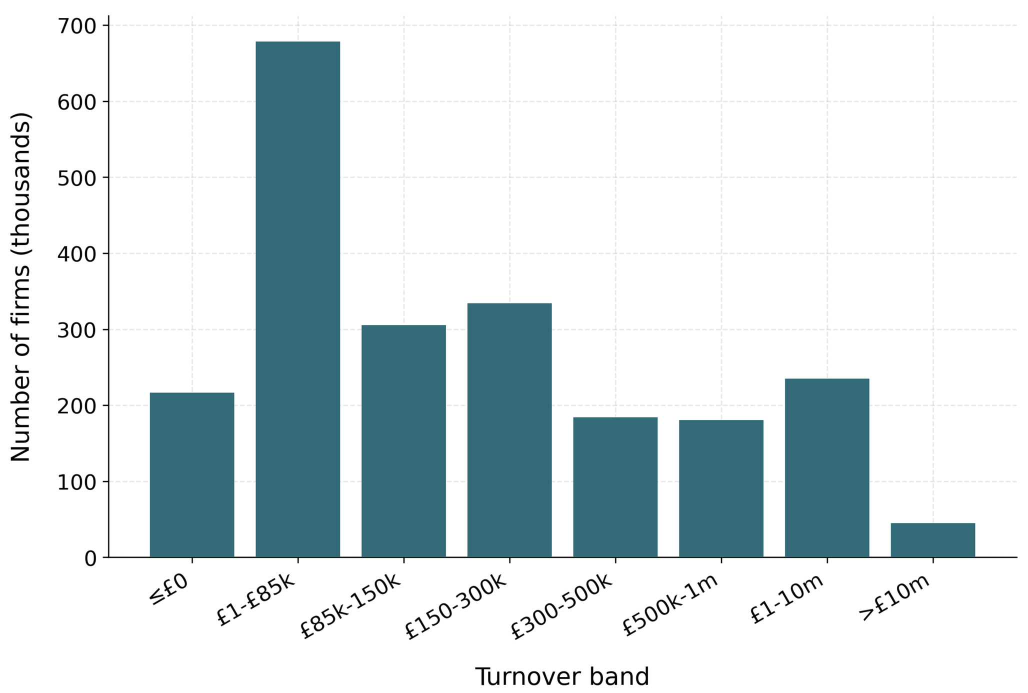
Costed on a *common static basis*:

- Level — move the threshold up or down.
- Shape — a graduated taper that phases VAT in.
- Rate — a reduced-rate band above the threshold.

Contribution

An open, reproducible firm-level microsimulation that:

1. costs level + shape + rate reforms on one static basis;
2. characterises the notch's distortion via its exact, elasticity-free dominated region;
3. shows via a placebo that synthetic bunching is mechanical;
4. adds a conditional behavioural layer.

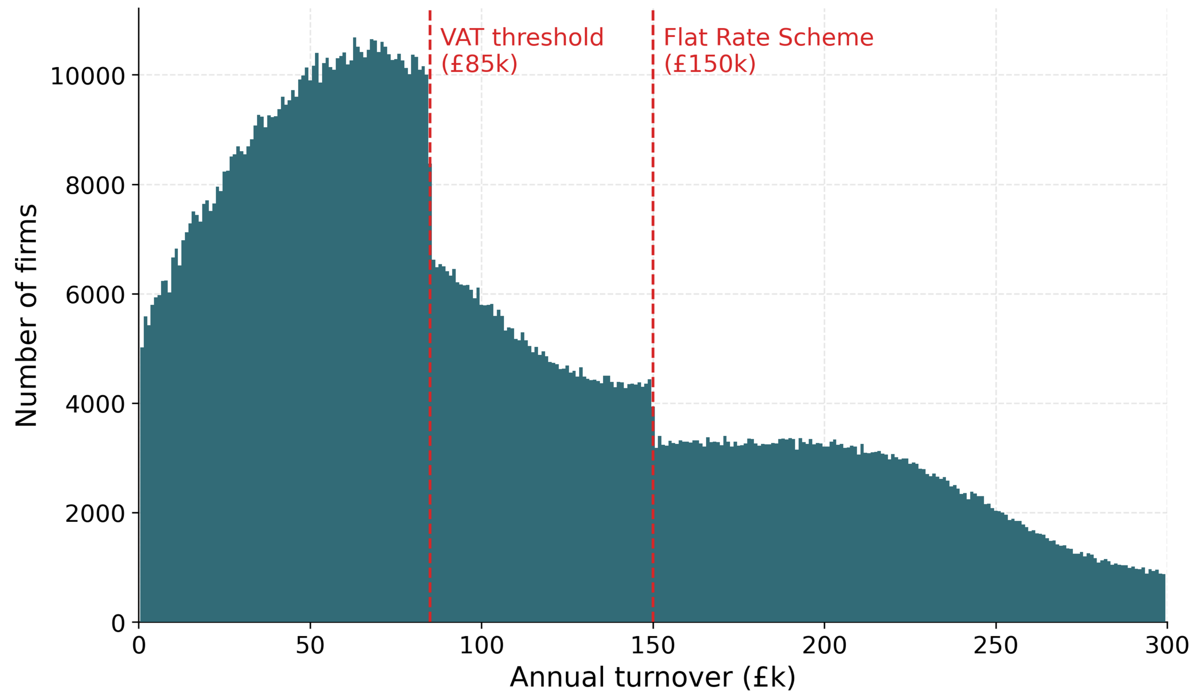


Method in one line

Draw synthetic firms by sector × turnover band; assign value added $v_i = y_i - x_i$. The threshold is a *generator parameter*, so data regenerate under any counterfactual. Firm weights $w_i = e^{\theta_i}$ are fit by gradient descent to HMRC + ONS aggregates.

Synthetic firm data

≈2.94m firm rows weighted to ≈2.5m UK firms, calibrated to HMRC + ONS aggregates (turnover-band counts, sector, employment, net VAT liability by band) at ≈90% accuracy. Calibrated *to* the aggregates, so agreement is an internal-consistency check, not external validation.

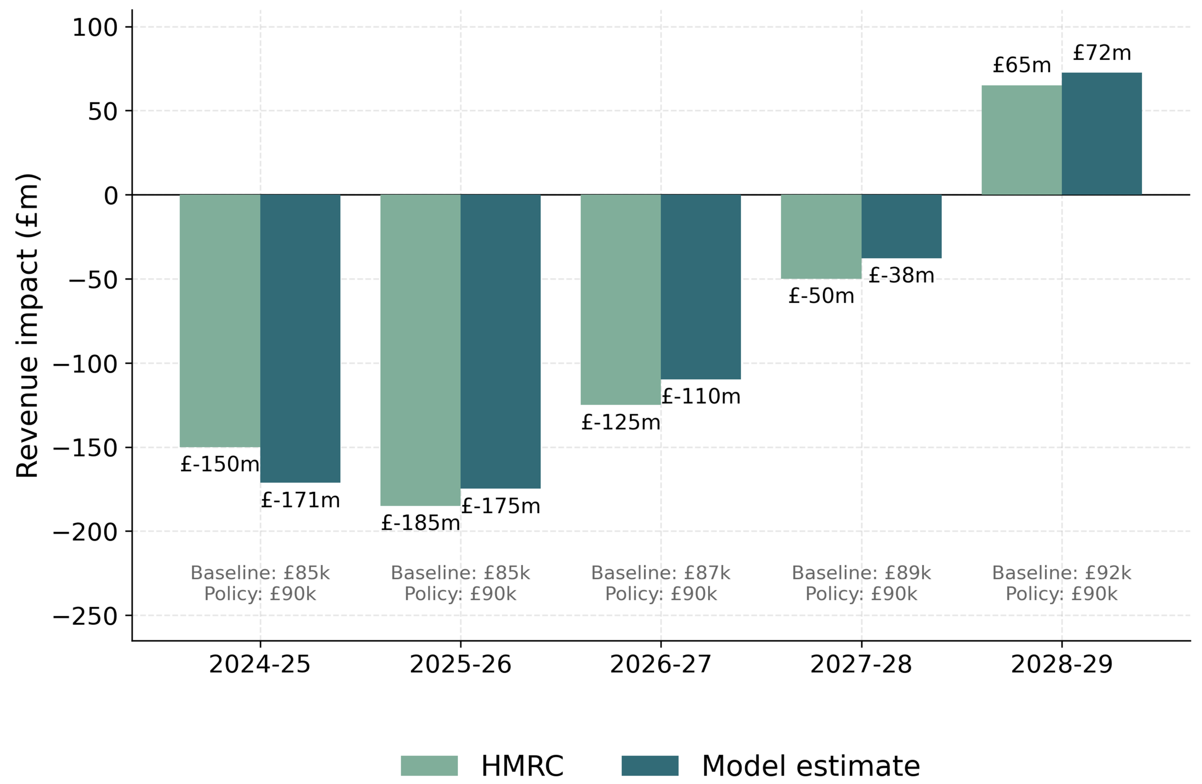


Static costing

$$R(T) = \sum_i w_i v_i \mathbb{1}\{y_i \geq T\}$$

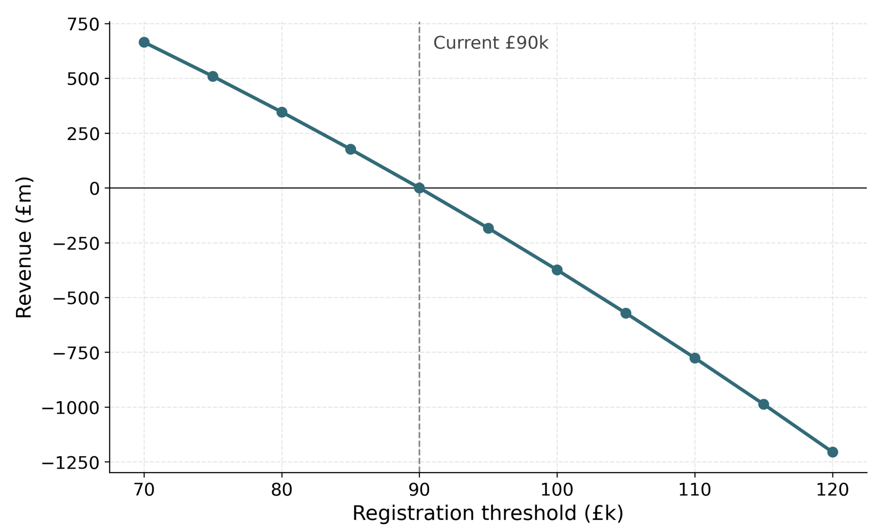
Revenue moves only through firms whose registration status flips.

Validation vs HMRC: the £85k→£90k April-2024 increase costs −£175m in 2025–26 vs HMRC's published −£185m — agreement within ∼£10m.



Threshold sweep (2025–26)

From a £90k base (£203.8bn), the relationship is roughly linear, ≈£150–220m per £5k step.

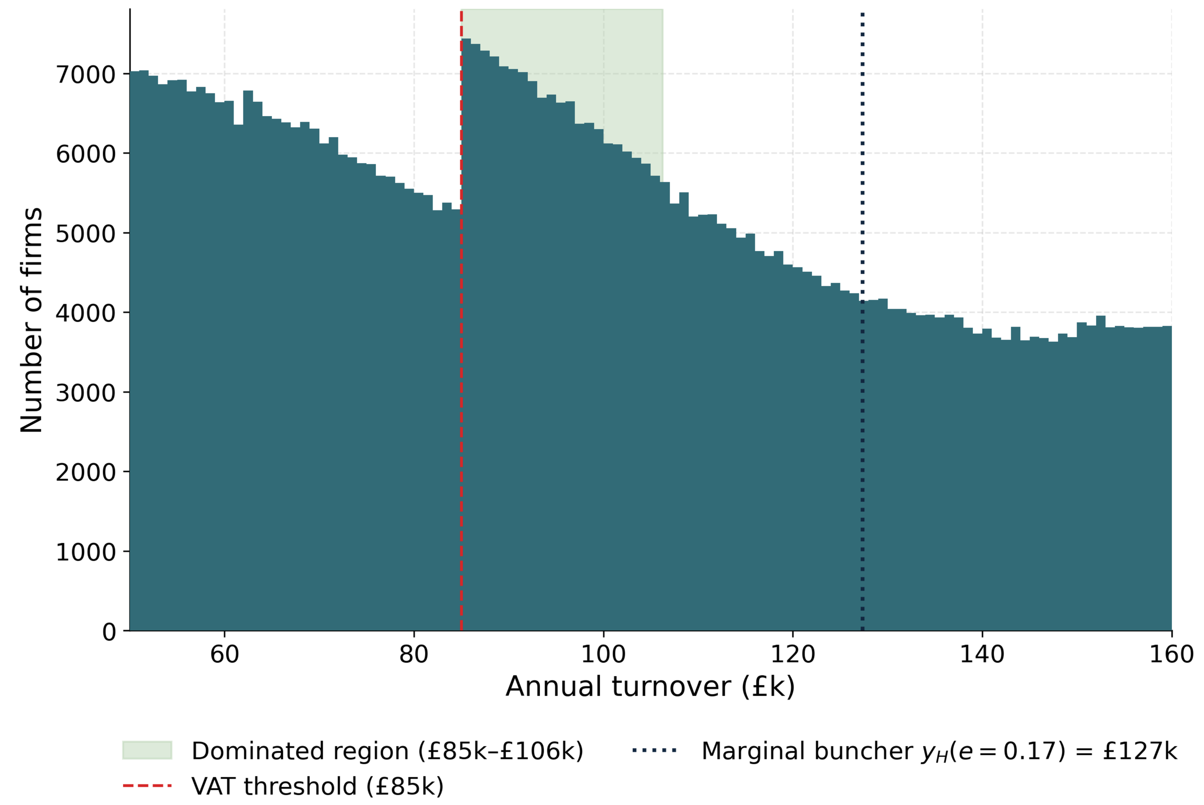


The dominated region

Profit jumps *down* by $\tau T^* = \pounds17{,}000$ at the threshold. The dominated region — turnover where no firm locates because it earns less than at T^* — has exact width

$$a = T^* \frac{\tau}{1-\tau} = \pounds21{,}250$$

(£85,000–£106,250), spanning ≈137,000 weighted firms. Exact and elasticity-free (conditional on the turnover-tax notch).

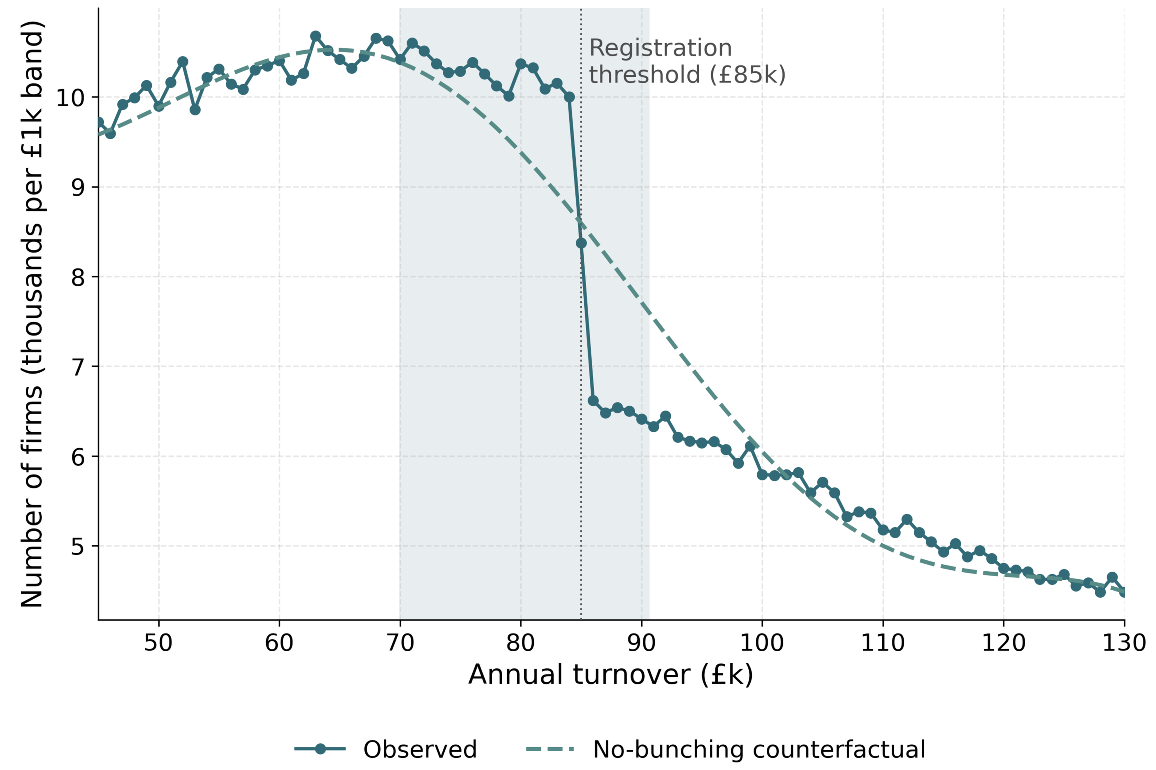


How reforms change it

- Raising the threshold relocates the £21,250 band unchanged.
- A banded reduced rate does not shrink it: reverting to 20% at the band top adds a *secondary* notch $a' = T_1 \frac{\tau - r}{1-\tau}$, so total width is £21,562 (15%) / £22,569 (10%) — it splits the band, not shrinks it.
- Only the graduated taper removes the distortion entirely.

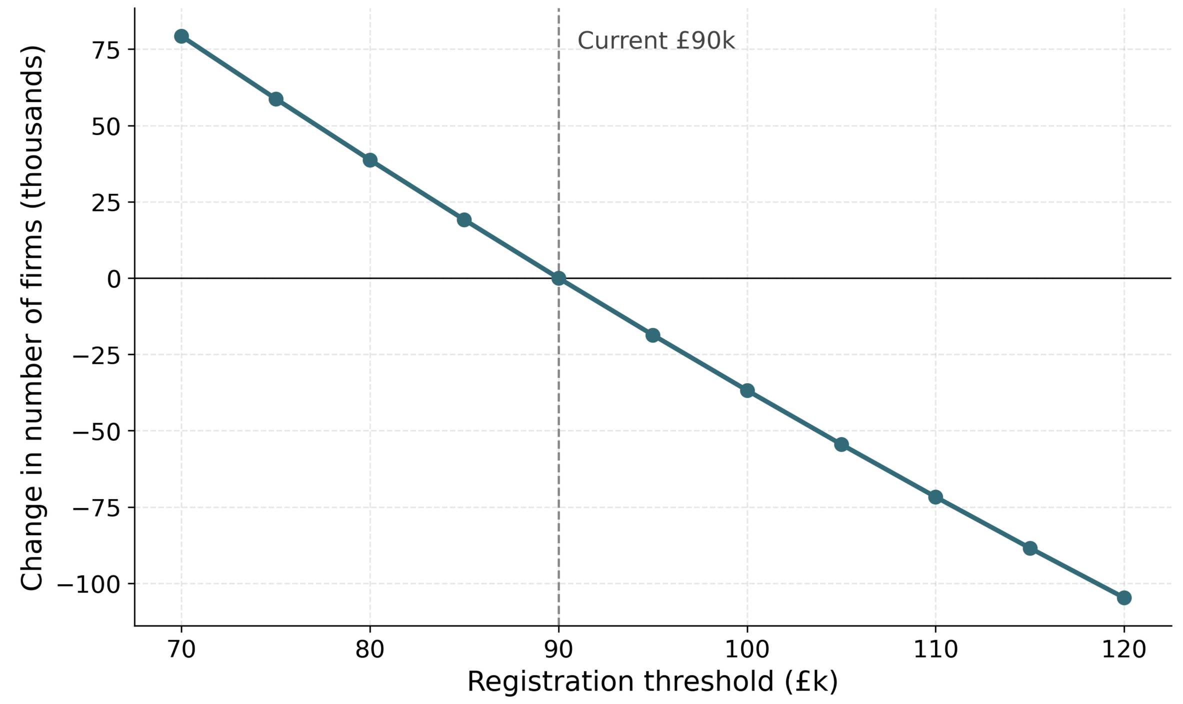
Bunching is mechanical

Estimator: $b = 0.060$, excess mass $E = 8{,}712$ firms. A placebo collapses E to 0–99 when the step is removed from the calibration target ⇒ bunching is inherited from coarse HMRC band targets, *not* firm behaviour. Behavioural bunching is therefore not identified on the synthetic data; the administrative estimate of Liu et al. (2021) remains the behavioural fact.



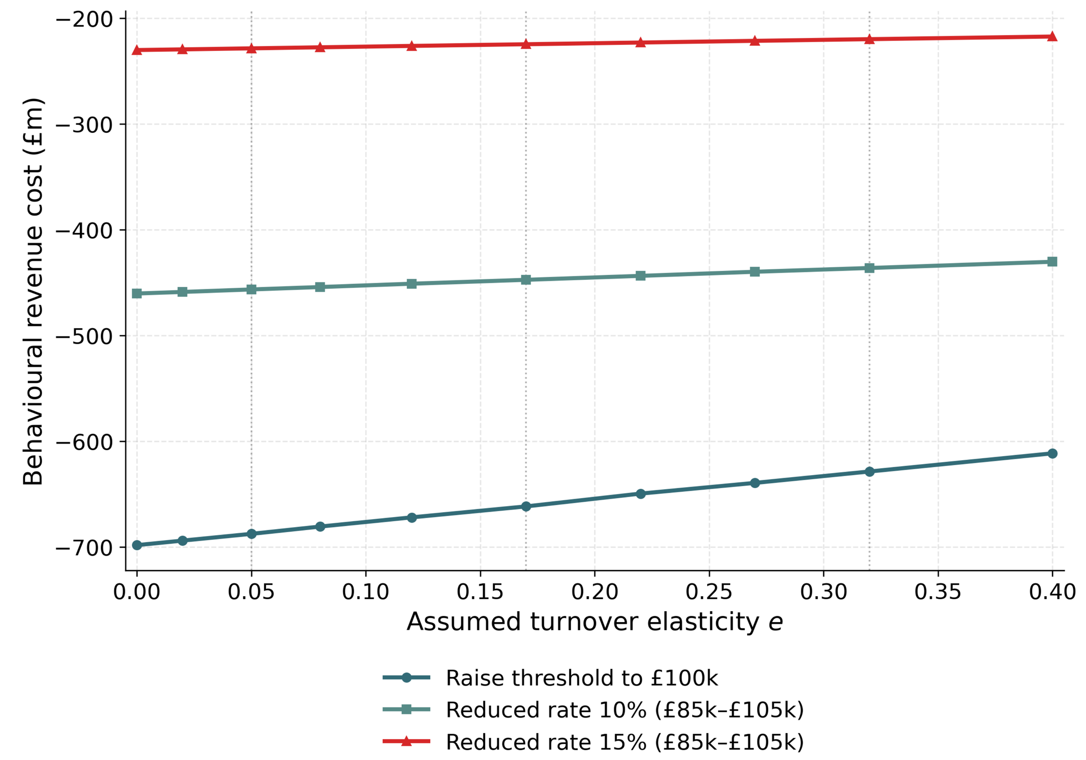
Reform menu (common £85k / £183.6bn base)

Reform	Type	Static	Dominated region
Raise to £100k	Location	−£508m	relocates, removes none
Taper £85–105k	Shape	−£336m	removed entirely
Reduced 10%	Rate	−£343m	splits (2nd notch)
Reduced 15%	Rate	−£171m	splits (2nd notch)



Behavioural layer (conditional)

Conditional on an assumed turnover elasticity e (*not* identified on synthetic data); sweep $e \in \{0.05, 0.17, 0.32\}$, midpoint 0.17. At $e = 0.17$: raise-to-£100k £508m→£292m; 10% band £343m→£273m; 15% band £171m→£135m, as firms broaden the base.



Conclusions

1. Prices level + shape + rate on one basis; reproduces HMRC's anchor costing.
2. On the dominated-region metric, only the taper removes the distortion — the level move is the dearest and least efficient.
3. Synthetic bunching is mechanical (a transparency lesson) — no elasticity is read off it.
4. Behavioural offsets are conditional on an assumed e; the static cost and dominated region are the elasticity-free anchors.

Future work & contact

Identify e from administrative micro-data; welfare / MVPF accounting; sectoral incidence.

Vahid Ahmadi — PolicyEngine
`github.com/PolicyEngine/firm-microsim-paper`